

Diabetes: stage the retina and assess the macula

Neovascularisation and macular involvement are separate questions in diabetic eye disease.

Two retinal dimensions

Stage the retinopathy and assess central risk

NPDR

Microvascular changes without new vessels.

PDR

New vessels; bleeding, traction and neovascular glaucoma risks.

Maculopathy

Oedema or ischaemia may accompany any stage.

Clinical meaning

Screening matters because retinal disease may be asymptomatic. Review duration and control of diabetes, BP and lipids. PRP reduces peripheral oxygen demand and VEGF drive; anti-VEGF can address macular fluid. OCT assesses thickness and fluid. A single central acuity measurement does not describe the full retinal burden.

Vascular occlusions: pallor versus haemorrhage

Arterial territory and venous drainage distribution help explain the fundus pattern.



CRAO: pale ischaemic retina contrasts with the red foveal centre.

Clinical meaning

Sectoral whitening suggests BRAO; congested veins and widespread blood-and-thunder haemorrhage suggest CRVO. A branch venous pattern is sectoral. Arterial loss or embolic transient monocular loss needs urgent systemic vascular assessment. In RVO, angiography evaluates ischaemia while OCT assesses macular fluid. Check BP and watch for neovascular complications.

Central distortion: distinguish degeneration from leakage

Drusen alone differ from an exudative macular lesion with haemorrhage or fluid.

Macular pattern

Finding	Clinical meaning
Drusen and pigment change	Dry AMD risk; deposits beneath RPE.
Fluid, haemorrhage or fibrosis	Assess active or previous wet AMD with CNV.
Contracting surface line on OCT	Epiretinal membrane.
Fluid lifting a retinal layer	Consider central serous chorioretinopathy.

Clinical meaning

New monocular Amsler distortion warrants earlier review. OCT shows structure and fluid; angiography assesses vascular leakage and CNV. Active wet AMD is treated with intravitreal anti-VEGF. Smoking cessation matters. Marked colour loss or RAPD also prompts optic nerve assessment rather than assuming that every central complaint is macular.

Flashes, floaters and a curtain: inspect the retina

PVD can be uncomplicated, but its symptoms can also announce a tear or detachment.

Mechanism of separation

Process	Key distinction
PVD	Posterior vitreous separates; the retina may remain attached.
Rhegmatogenous RD	Liquefied vitreous passes through a retinal break.
Tractional RD	A contracting membrane pulls retina away.
Exudative RD	Subretinal leakage without a break or traction.

Clinical meaning

A Weiss ring supports PVD, but the peripheral retina still needs examination. Blood obscuring the fundus may require B-scan. A break with attached retina can receive retinopexy; established detachment needs reattachment management. Refer acutely symptomatic detachment urgently, especially while the macula remains attached. Explain new symptoms as urgent return triggers.